FINANCIAL MARKETS ANALYTICS

Dynamic Asset Allocation

Variant B

Classical Markowitz · Michaud Resampling
Ledoit-Wolf · Hierarchical Risk Parity

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ORAL PRESENTATION

RESEARCH QUESTION

Can sophistication beat simple allocation rules?

Do optimized portfolios improve realized risk-adjusted performance after estimation error, turnover, and costs?

01

Performance

Return · volatility
Sharpe · drawdown

02

Robustness

Sampling error
Covariance stability

03

Implementability

Concentration · turnover
Transaction costs

Benchmarks matter: added complexity must earn its place out of sample.

INVESTMENT UNIVERSE

23 ETFs preserve crisis history without backfilling

CORE UNIVERSE

23 liquid, diversified market exposures



- Equities 9
- Government bonds 5
- Credit 3
- Commodities 3
- Real estate 2
- Cash 1

WHY ETFs?

- Broad exposure without stock-selection noise
- Transparent economic roles across asset classes
- Tradable instruments with interpretable weights

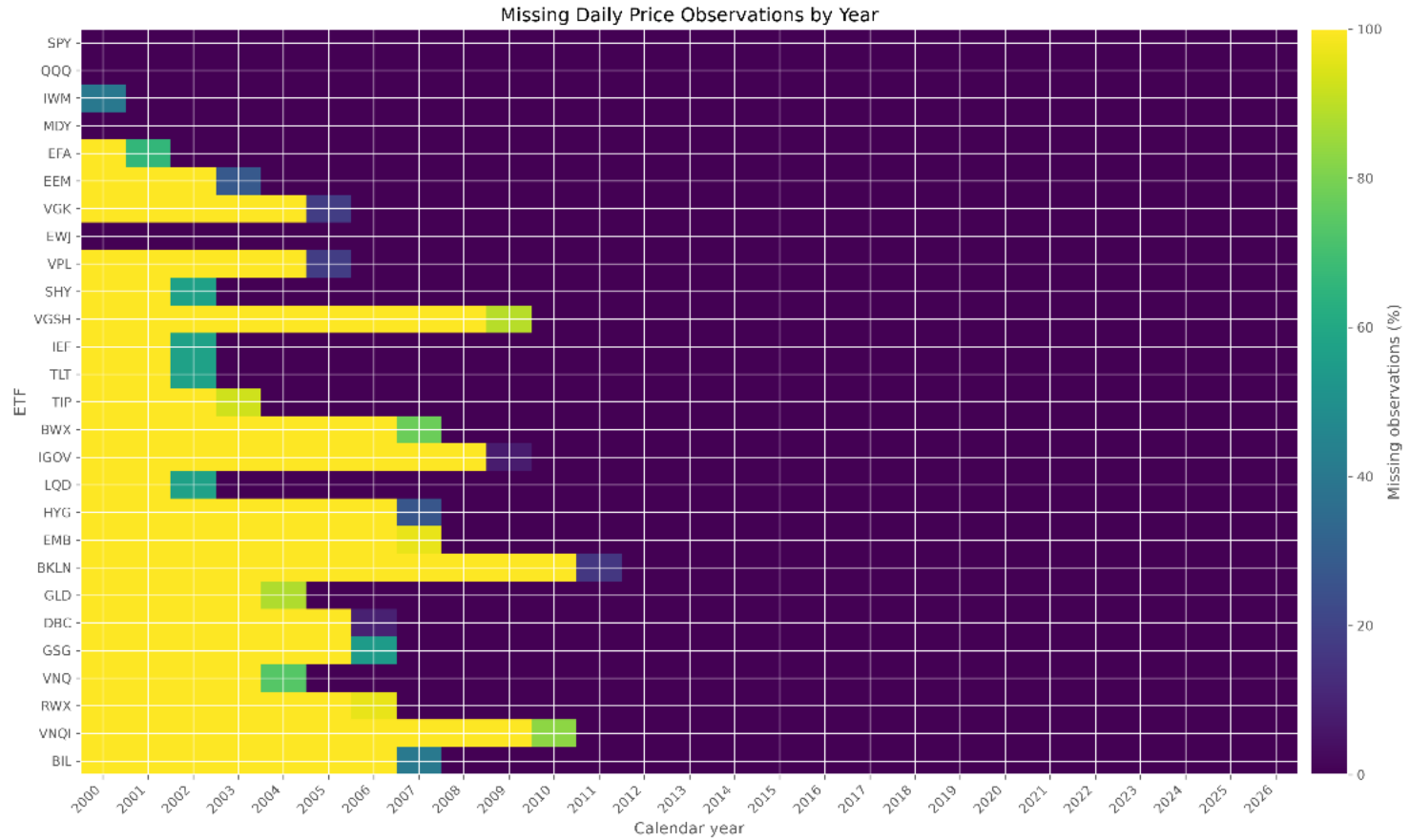
WHY 23?

Retain the 2008 crisis while excluding four late-inception ETFs:

IGOV · VGSH · VNQI · BKLN

Same assets for every final strategy

Structural missingness defines the analytical sample



Source: outputs/figures/missing_values_heatmap.png

DATA

- Yahoo Finance adjusted ETF prices
- Daily observations → month-end prices
- January 2008 → May 2026 core panel

CLEANING RULE

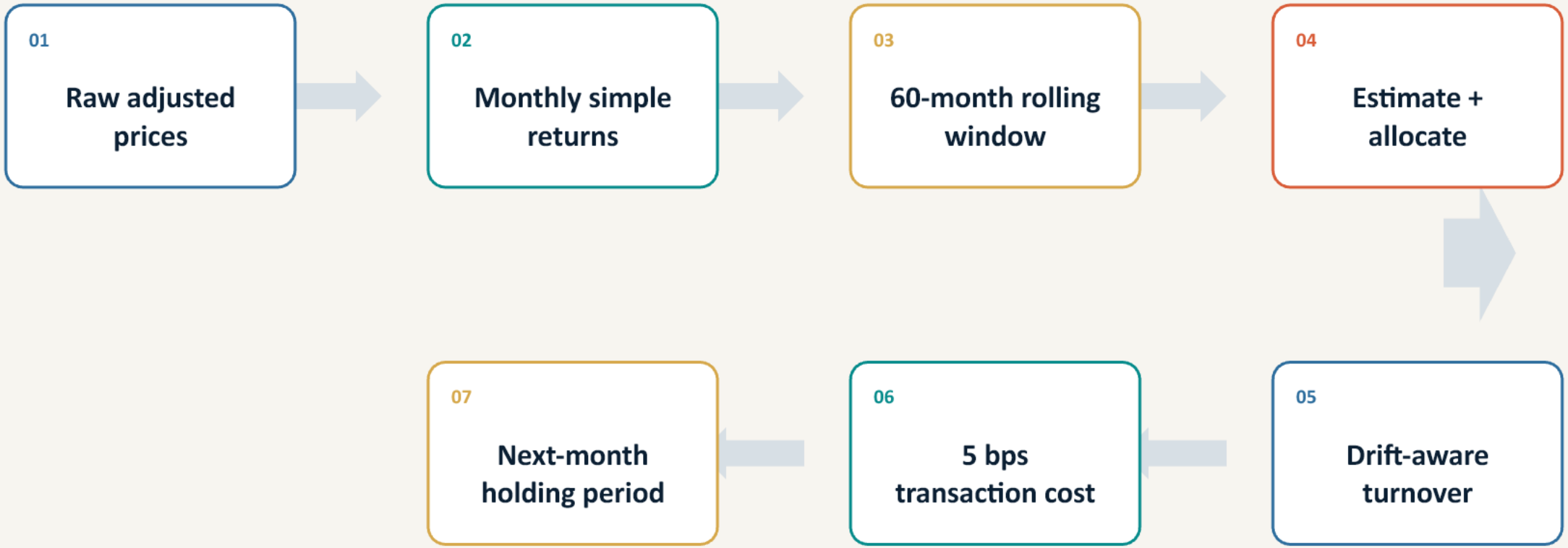
No interpolation

No backfilling

No synthetic prices

Missing blocks are pre-inception history—not zero returns.

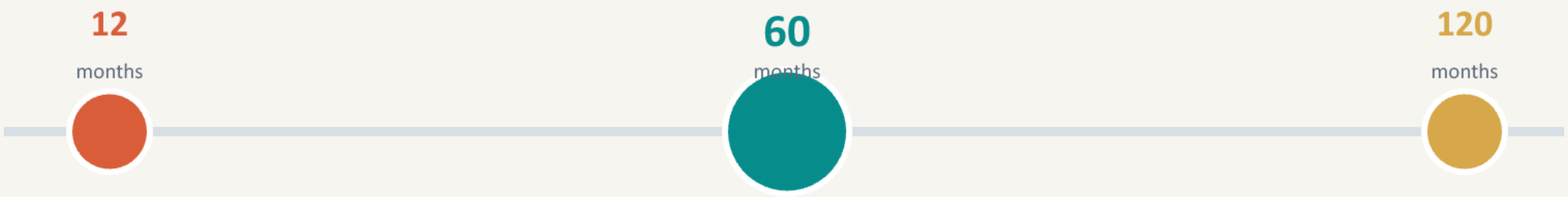
The portfolio is rebuilt using only available information



Repeat monthly · identical calendar for benchmarks and optimized portfolios

WHY THIS ROLLING WINDOW?

60 months balances stability with adaptability



Too reactive

Noisy means and correlations

Chosen balance

Five years of data with regime adaptation

Too slow

Stable estimates, but stale regimes

← ADAPTABILITY INCREASES

STABILITY INCREASES →

CLASSICAL OPTIMIZATION

Constraints make Markowitz implementable

THREE OBJECTIVES

Minimum Variance

$$\min \; w^T \Sigma w$$

Maximum Sharpe

$$\max \; (w^T\mu - r^f) / \sqrt{(w^T\Sigma w)}$$

Target Volatility

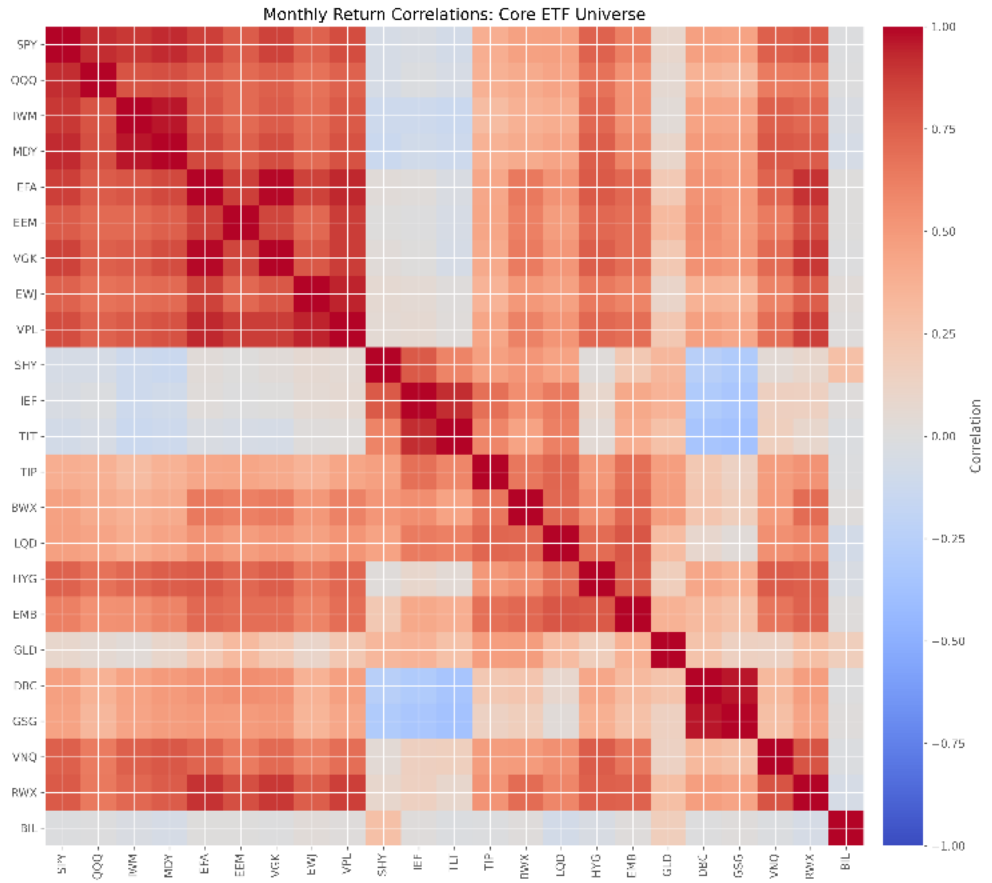
$$\max \; w^T\mu \quad \text{subject to} \quad \sigma \leq 10\%$$

IMPLEMENTATION CONSTRAINTS

Σ weights = 1	Fully invested
$0\% \leq w_i \leq 25\%$	Long-only + concentration cap
No leverage	Comparable and investable
Monthly	Matches data frequency
5 bps	Penalizes unstable turnover

Why 25%? Limit estimation-driven concentration without forcing equal weights.

Shrinkage stabilizes the risk model before optimization



23 assets × 60 observations

Sample covariance

Direct historical estimate

- Noisy off-diagonal terms
- Poor conditioning can amplify weight changes

$$\Sigma^{LW} = \delta F + (1 - \delta)S$$

structured target + sample matrix

Small bias → lower estimation variance → better conditioning

Michaud averages portfolios across plausible inputs

01 **Estimate μ and Σ**

02 **Simulate M universes**

03 **Optimize each**

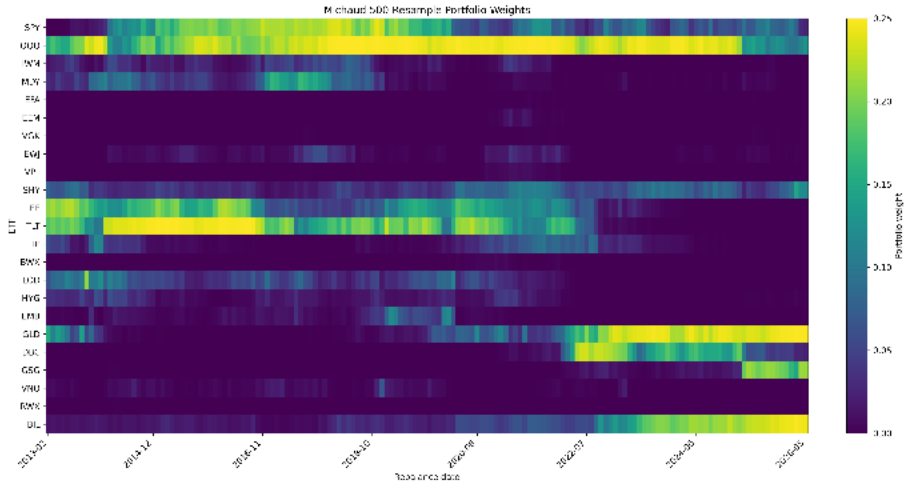
04 **Average weights**

AVERAGE THE DECISIONS

$$\bar{w} = (1/M)\,\Sigma_m\, w_m$$

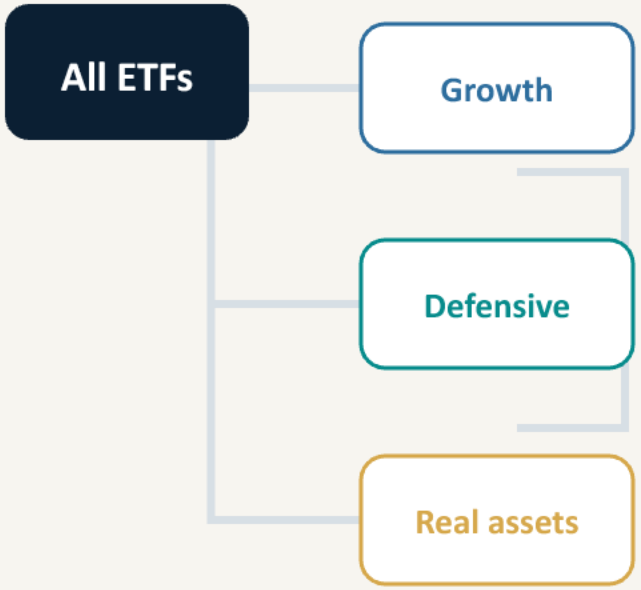
100 · 250 · 500 nested resamples

Why? Reduce dependence on one noisy efficient-frontier estimate.



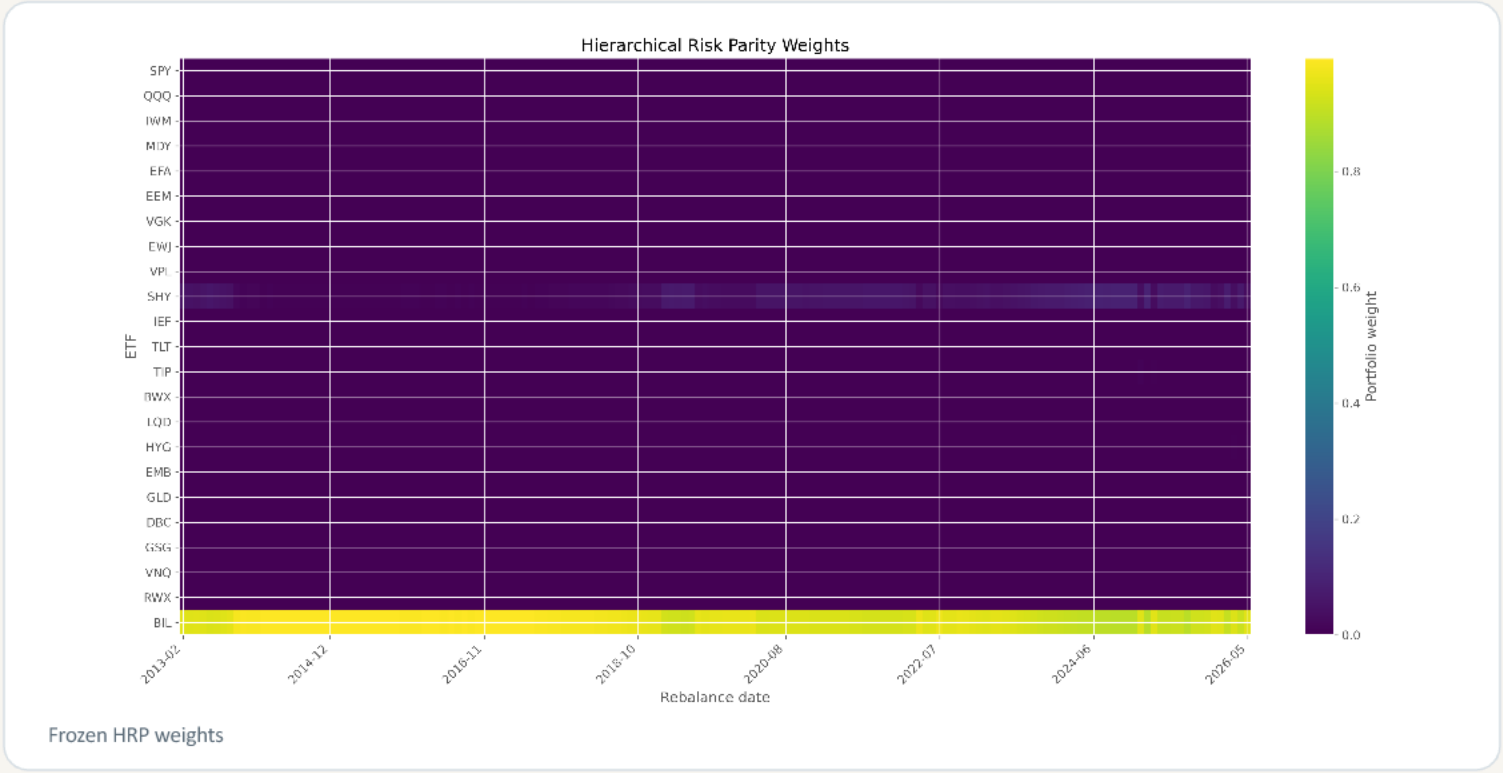
Frozen output: Michaud 500 rolling allocation

HRP removes inversion—but not concentration risk



Correlation distance
→ clustering
→ recursive bisection

NO COVARIANCE INVERSION



Frozen HRP weights

Empirical result: near-zero-volatility BIL dominated the unconstrained recursive allocation.

WALK-FORWARD BACKTEST

Every reported return is genuinely out of sample

60 monthly observations

t−60 — historical returns only — t−1

→ **t** estimate + set weights → **t+1** realized return

No look-ahead bias — The return earned after the decision never enters its own estimation window.

COMMON FINAL PERIOD

31 Mar 2013 → 159 identical months → **31 May 2026**

SAME UNIVERSE · SAME DATES · SAME COSTS

FINAL RESULTS

No single strategy wins every objective

Strategy	Return	Volatility	Sharpe	Max DD	Turnover	Eff. assets
Equal Weight	6.1%	9.1%	0.476	-19.1%	13.2%	23.00
60/40	7.0%	9.8%	0.540	-23.3%	12.5%	13.89
Min Variance Sample	2.2%	2.8%	0.096	-8.0%	26.1%	4.71
Min Variance Ledoit-Wolf	2.3%	2.9%	0.114	-9.1%	23.5%	5.35
Max Sharpe Sample	9.2%	7.9%	0.909	-20.7%	99.1%	4.50
Max Sharpe Ledoit-Wolf	9.9%	8.5%	0.922	-22.0%	88.1%	4.77
Target Volatility Sample	11.3%	11.5%	0.822	-22.6%	103.9%	4.50
Target Volatility Ledoit-Wolf	11.4%	11.6%	0.818	-22.3%	98.7%	4.52
Michaud 100	9.2%	8.0%	0.903	-20.5%	91.5%	6.23
Michaud 250	9.2%	8.0%	0.899	-20.5%	81.5%	6.24
Michaud 500	9.2%	8.0%	0.900	-20.5%	78.0%	6.25
HRP	1.6%	0.6%	-0.670	-0.3%	7.5%	1.09

BEST SHARPE

Max Sharpe LW 0.922

HIGHEST RETURN

Target Vol LW 11.38%

LOWEST TURNOVER

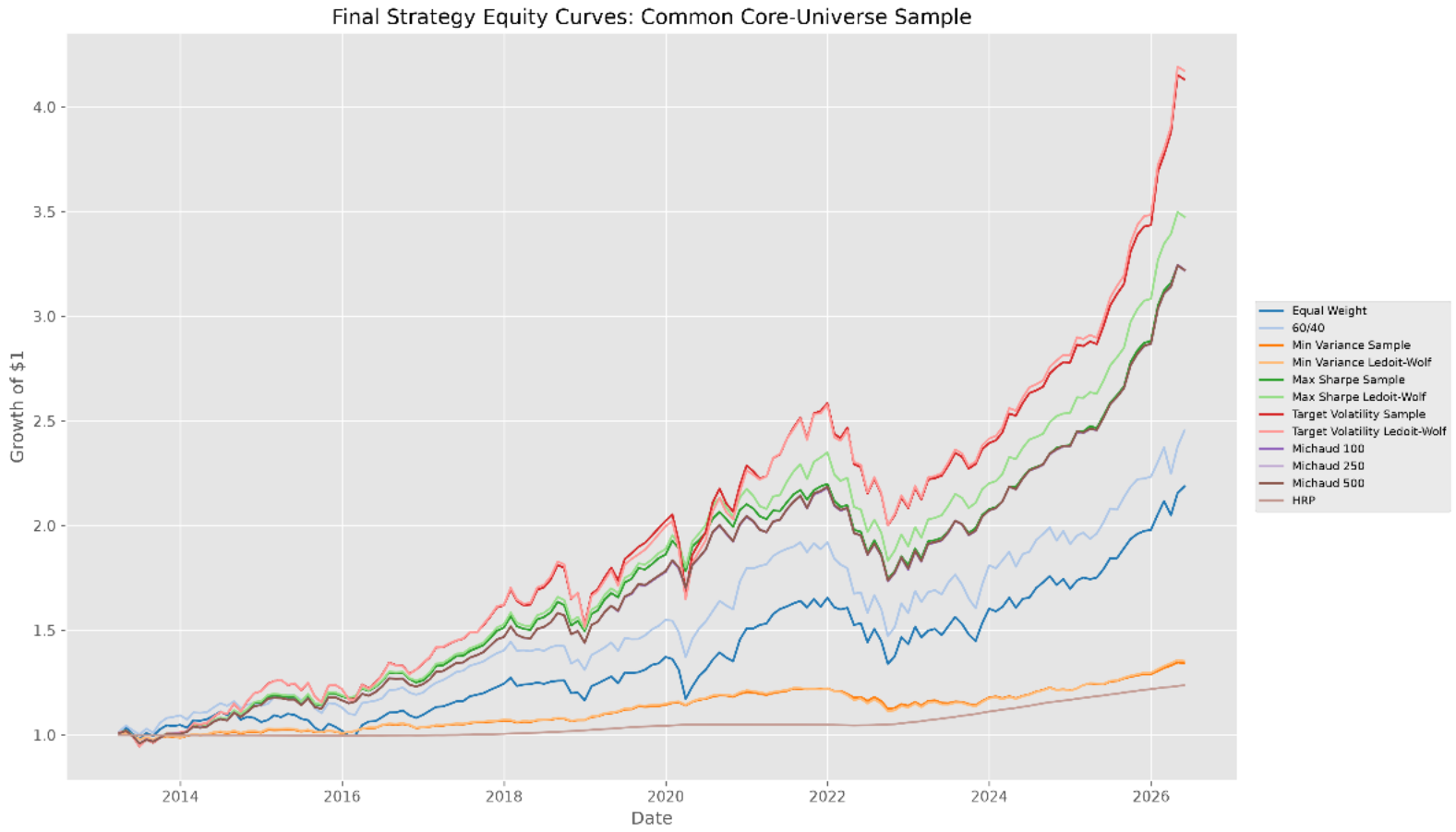
HRP 7.49%

MOST DIVERSIFIED

Equal Weight 23.00

CUMULATIVE PERFORMANCE

Terminal wealth reflects different amounts of risk



Source: outputs/figures/final_equity_curves.png

READ THE PATH

1 Target Volatility finishes highest by accepting more risk.

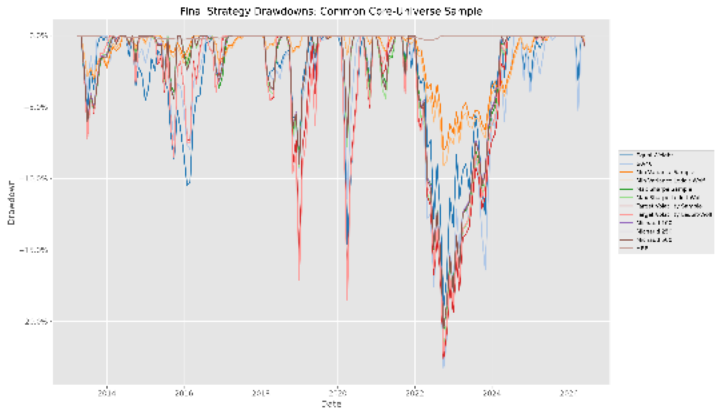
2 Maximum Sharpe and Michaud deliver the strongest risk-adjusted middle path.

3 HRP behaves like cash after concentrating in BIL.

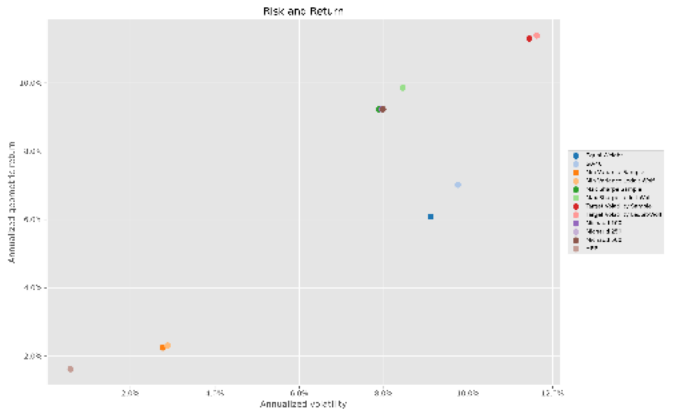
Highest wealth ≠ universal best

Risk, turnover, and diversification tell different stories

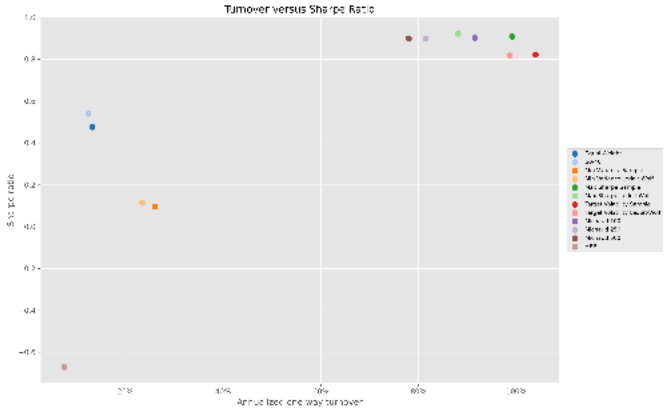
DRAWDOWNS



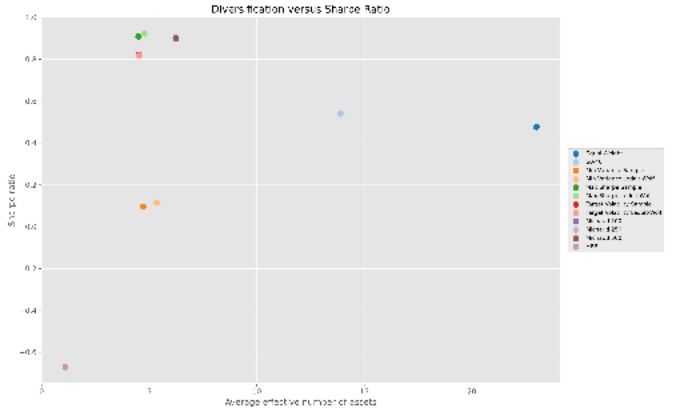
RISK–RETURN



TURNOVER–SHARPE



DIVERSIFICATION–SHARPE



INTERPRETATION

Robustness mattered as much as optimization

Maximum Sharpe + Ledoit-Wolf	**Best realized Sharpe**	The return signal added value after the covariance input was stabilized.
Target Volatility	**Highest terminal wealth**	The gain came with more volatility—not a free improvement.
Michaud 500	**Best optimized diversification**	A small Sharpe sacrifice bought lower turnover and broader weights.
HRP	**Defensive collapse into BIL**	No inversion did not prevent extreme concentration in a cash-like asset.

Method × estimator × universe design determines the realized portfolio.

LIMITATIONS

The evidence is informative—not definitive

01 Fixed ETF universe

Selection and survivorship bias remain possible.

02 Yahoo Finance

Reproducible, but not an institutional total-return source.

03 One realized history

Small ranking differences lack formal structural proof.

04 Constant transaction costs

No time-varying spreads, impact, or liquidity effects.

05 Taxes and investor frictions

Taxes, implementation constraints, and mandates are omitted.

06 Distribution assumptions

Michaud normality may understate tails and regime shifts.

Interpret the results as comparative evidence—not universal dominance.

FUTURE WORK

Robustness checks should challenge the design choices

Black-Litterman
01
Structured priors and investor views

Regime models
02
Time-varying risk structure

Alternative windows
03
36 / 60 / 120 and quarterly

Constrained HRP
04
No cash · caps · asset-class floors

Implementation
05
Dynamic costs + extended universe

Principle **Pre-specify robustness tests—do not search ex post for the best-looking result.**

CONCLUSION

Complexity helped only when paired with discipline

01	**Walk-forward design**	Same universe, dates, costs, and next-month timing
02	**Best Sharpe**	Maximum Sharpe Ledoit-Wolf · 0.922
03	**Highest wealth**	Target Volatility Ledoit-Wolf · more risk
04	**Best robustness trade-off**	Michaud 500 · broader weights and lower turnover

Robust portfolio design is more than choosing an optimizer.